AGE 310: Instrumentation and Measurements

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Abstract

Lecture 14 covers Machine Vision, the core technology that enables robots to understand and interact with the complex agricultural environment. We examine the EM Spectrum as a sensing tool, detail the Image Processing Pipeline (from acquisition to classification), and analyze practical applications such as automated weed detection, selective spraying (*See and Spray*), and post-harvest quality grading.

Lecture 14: Machine Vision in Agriculture

14.1 Fundamentals of Machine Vision

14.1.1 Definition and Components

Machine Vision (MV) is the discipline used to provide imaging-based inspection, analysis, and guidance for automated processes. It links optics, electronics, and algorithms.

Machine Vision System Components:

1. **Illumination:** Provides controlled light (e.g., specific LED wavelengths) to highlight features and minimize shadows. This is often the most critical component for field work.
2. **Image Acquisition:** The camera (lens and sensor) converts photons into an electrical signal.
3. **Processing Unit:** A high-speed computer (CPU/GPU) runs the software algorithms.
4. **Actuator Interface:** Converts the visual decision (e.g., "Weed detected at X, Y") into a physical command (e.g., open nozzle 5).

14.1.2 Digital Image Basics

A digital image is a two-dimensional grid (*matrix*) of picture elements, or **pixels**.

- **Pixel Value:** The intensity or brightness value at that coordinate.
- **RGB Image:** Each pixel has three values (Red, Green, Blue), typically ranging from 0 to 255 (8-bit).
- **Hyperspectral Image:** Captures hundreds of narrow, continuous wavelength bands, providing detailed chemical composition data (*e.g.*, sugar content).

14.2 The Image Processing Pipeline

The conversion of raw image data into a usable decision involves a series of standardized steps:

14.2.1 1. Image Acquisition and Pre-processing

- **Acquisition:** Capturing the image, often synchronized with the tractor's speed.
- **Filtering (LPF):** Applying filters (like Gaussian Blur) to reduce random noise without losing edge information.
- **Color Space Conversion:** Converting RGB to Grayscale, HSV (Hue, Saturation, Value), or specialized indices (e.g., ExG) for easier analysis.

14.2.2 2. Image Segmentation

The process of dividing the image into meaningful regions, separating the foreground (*object of interest*) from the background.

Thresholding (Simple Segmentation): The simplest method involves setting a cut-off value based on intensity or color index.

Excess Green Index (ExG): Exploits the high Green reflection of plants against soil.

$$\text{ExG} = 2G - R - B$$

If ExG is above a specific threshold, the pixel is classified as "Plant." The output is a **Binary Mask** (black for soil, white for plant).

14.2.3 3. Feature Extraction and Classification

Once segmented, the system measures physical characteristics of the identified objects (*blobs*).

Feature Examples:

- **Shape (Morphology):** Area (pixel count), perimeter, circularity, and aspect ratio.
- **Color/Texture:** Average HSV values, smoothness, or coarseness (used to detect disease spots).

Classification (Decision):

- **Rule-Based:** Uses simple logic based on features. (IF Area $> X$ AND Circularity $< Y \rightarrow$ Weed)
- **Machine Learning (Deep Learning):** Uses algorithms (CNNs) trained on thousands of examples to recognize complex patterns (e.g., distinguishing maize from *amarantus* weed).

14.3 Applications in Agricultural Automation

14.3.1 Selective Spraying (See and Spray)

This is the most impactful commercial application of MV.

- **Green-on-Brown:** Used in fallow or post-harvest fields. The camera identifies any green object on the brown soil.
- **Green-on-Green:** The challenge is distinguishing the crop from the weed (e.g., small weeds amongst young beans). This requires AI models trained on shape and texture differences.
- **Action:** Once classified as a weed, the system sends an instant ON command to a micro-solenoid nozzle (Lecture 7) directly above the weed, achieving high chemical savings.

14.3.2 Post-Harvest Grading and Quality Control

Used in packing houses to automate sorting, ensuring export quality standards are met.

- **Function:** Apples/Mangos pass under high-speed cameras.
- **Grading Criteria:**
  1. *Size:* Calculated from the pixel area.
  2. *Color/Ripeness:* Sorting based on HSV values (e.g., Hue for maturity).
  3. *Defects:* Detecting bruises, mold, or insect damage (Texture analysis).
- **Actuation:** Air jets or mechanical pushers direct the fruit to the correct bin.

14.3.3 Disease and Pest Monitoring

MV enables early and accurate identification of biotic stresses.

- **Mobile Applications (e.g., Nuru for Cassava):** Uses smartphone cameras and pre-trained AI models to identify specific leaf symptoms (e.g., Cassava Mosaic Disease) and provide instant recommendations to the farmer.
- **Drone Scouting (Lecture 13):** High-resolution cameras detect subtle discoloration or wilting in the early stages of disease spread, allowing targeted intervention.

Calculation Example 1: Weed Area and Chemical Usage

A machine vision system identifies a weed blob consisting of 500 pixels. The camera system is calibrated such that 1 pixel $= 1$ mm^2 on the ground. The spot sprayer applies 1 µL of chemical per 100 mm^2 of weed area.

1. Calculate the physical area of the weed in cm^2.
2. Calculate the volume of chemical applied to this single weed in µL.

Solution:

1. **Physical Area (A_{mm^2}):** 500 pixels $\times$ 1 mm^2/pixel $= 500$ mm^2. **Area in cm^2:** 1 cm^2 $=$ 100 mm^2.
$$A_{\text{cm}^2} = 500 \text{ mm}^2 / 100 \text{ mm}^2/\text{cm}^2 = 5 \text{ cm}^2$$

2. **Chemical Volume (V):** The area is 5 cm^2, or 500 mm^2. Required chemical volume: $\frac{1\ \mu\text{L}}{100\ \text{mm}^2} \times 500\ \text{mm}^2 = 5\ \mu\text{L}$

This precision shows the fundamental economic benefit of MV in saving expensive chemicals.

14.4 Field Challenges and Future Outlook

14.4.1 Key Challenges in the Field

- **Illumination Variation:** The uncontrolled field lighting (sun, clouds, shadows) is highly detrimental to color-based algorithms. Requires active lighting enclosures.
- **Occlusion:** Overlapping leaves or crops hiding the target object from the camera. This is particularly difficult for harvesting robots.
- **Real-Time Processing:** Systems must process images (e.g., 30 frames$/$s) instantly to control fast-moving actuators (tractor speed ~ 2 m$/$s). Requires powerful embedded GPUs (e.g., NVIDIA Jetson).

14.4.2 Future of Ag Vision

The trend is moving towards combining multiple spectral sensors (RGB $+$ NIR $+$ Thermal) and utilizing advanced AI models (YOLOv8, Mask R-CNN) that can achieve complex *Green-on-Green* segmentation and object tracking, making automation more reliable.

14.5 Summary of Key Terms (Swahili/Context)

- **Machine Vision (MV):** Maono ya Mashine (Automated image analysis).
- **Pixel:** Nukta Picha (Basic element of a digital image).
- **Segmentation:** Kugawanya Picha (Separating foreground from background).
- **Binary Mask:** Barakoa ya Mbili (Black and white image separating objects).
- **ExG:** Kielezo cha Kijani Kingi (Excess Green Index).
- **Occlusion:** Kufichwa (Target object being hidden by overlap).
- **Hyperspectral:** Masafa Mengi (Capturing narrow, continuous light bands).
- **See and Spray:** Ona na Nyunyizia (Selective spraying technique).

Lecture 14: Review Questions

SECTION A: Basic Concepts (5 Questions)

1. Define *Machine Vision*. Why is *Illumination* considered the most critical component in a field MV system?
2. Explain the process of *Segmentation* in image processing. What is the output product of a successful segmentation process?
3. State the formula for the *Excess Green Index* (ExG). How is this index used to differentiate crop from soil?
4. Differentiate between *Rule-Based Classification* and *Machine Learning Classification* in the context of identifying a diseased leaf.
5. In post-harvest quality control, list two criteria used by MV systems to grade apples.

SECTION B: Advanced Analysis and Calculations (10 Questions)

6. **ExG and Classification Threshold:** A camera detects a patch of green grass reflecting $\mathrm{R} = 50, \mathrm{G} = 150, \mathrm{B} = 75$. Calculate the ExG value for this pixel. If the system uses a segmentation threshold of $\mathrm{ExG} > 100$, will this pixel be classified as a plant or background?
7. **Real-Time Processing Requirement:** A camera captures images at 30 frames$/$s. The tractor travels at 1.5 m$/$s. Calculate the physical distance (in cm) the tractor moves between two consecutive frames. Explain the challenge this poses for the Actuator (e.g., a spraying nozzle).
8. **Hyperspectral vs. RGB:** Explain the fundamental difference between *RGB* and *Hyperspectral* imaging. Provide one specific application (other than NDVI) where hyperspectral data is essential for agricultural analysis.
9. **Weed Identification (Feature Extraction):** A system is trained to distinguish round *Purslane* weeds from the long, narrow leaves of *Maize*. Which morphological feature (Area, Perimeter, or *Aspect Ratio*) would be the most effective feature to extract for classification? Justify your choice.
10. **Occlusion Challenge and Mitigation:** Define *Occlusion* in the context of robotic fruit picking. What specialized sensing technique (e.g., stereo vision or LiDAR) is required to determine the 3D position of a partially hidden object?
11. **Image Storage Requirement:** A farmer collects $1,000$ high-resolution aerial images, each 12 Megapixels (MP) at 3 bytes per pixel (RGB). Calculate the total storage space required in Gigabytes (GB). (1 GB $= 1024$ MB).
12. **Green-on-Green Challenge:** Explain why *Green-on-Green* weed detection is significantly more difficult than *Green-on-Brown* detection, requiring the use of advanced AI (Deep Learning).
13. **Illumination Compensation:** Explain why commercial *See and Spray* systems often utilize a controlled, enclosed lighting source (LEDs) rather than relying solely on natural sunlight when operating in the field.

14. **Sensor Selection for Quality:** A processing plant needs to detect small, dark fungal spots on harvested cotton bolls. Would you recommend a standard visible RGB camera or a specialized UV (Ultraviolet) camera? Justify your choice based on fungal properties.

15. **Actuator Speed Requirement:** The vision system detects a weed and needs to trigger a nozzle for 50 ms (milliseconds). If the tractor speed is 2 m/s, calculate the length of the field sprayed by that nozzle during the ON time, and explain the required speed of the Actuator itself (Solenoid Valve).